BY ORDER OF THE COMMANDER
AIR FORCE MATERIEL COMMAND

AIR FORCE MATERIEL COMMAND
INSTRUCTION 21-102

10 DECEMBER 2025

Maintenance

ANALYTICAL CONDITION INSPECTION (ACI) PROGRAMS

COMPLIANCE WITH THIS PUBLICATION IS MANDATORY

ACCESSIBILITY: Publications and forms are available for downloading or ordering on the e-publishing website at www.e-publishing.af.mil.

RELEASABILITY: There are no restrictions on releasing this publication.

OPR: HQ AFMC/A4/10-EN
Certified by: HQ AFMC/A4/10
Supersedes: AFMCI 21-102, 20 November 2019
Pages: 13

This instruction implements DAFPD 21-1, *Maintenance of Military Materiel*. It provides guidance and procedures for establishing and monitoring Analytical Condition Inspection (ACI) programs for aerospace equipment. This publication applies to all AFMC units of the Regular Air Force. This instruction does not apply to the Air Force Reserve, Air National Guard or the United States Space Force. Ensure that all records created as a result of processes prescribed in this publication adhere to AFI 33-322, *Records Management and Information Governance Program*, and are disposed in accordance with the Air Force Records Disposition Schedule, which is located in the Air Force Records Information Management System. Refer recommended changes and questions about this publication to the OPR using the DAF Form 847, *Recommendation for Change of Publication*; route DAF Forms 847 from the field through the appropriate functional chain of command. This AFMCI may be supplemented at any level. The authorities to waive wing/unit level requirements in this publication are identified with a Tier ("T-0, T-1, T-2, T-3") number following the compliance statement. See DAFMAN 90-161, *Publishing Processes and Procedures*, for a description of the authorities associated with the Tier numbers. Submit requests for waivers through the chain of command to the appropriate Tier waiver approval authority, or alternately, to the Publication OPR for non-tiered compliance items. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Air Force.

SUMMARY OF CHANGES

This document has been revised and should be completely reviewed. Changes include replacing the ACI Sample Sizes table with a tool, the numbering of ACI Work Specifications, reference updates and general cleanup.

1. Overview..... 2
2. Responsibilities. 2
3. Procedures. 3
4. Initiating Analytical Condition Inspection (ACI) Programs. 4
5. Analytical Condition Inspection (ACI) Sampling Plans. 5
6. Analytical Condition Inspection (ACI) Samples. 5
7. Analytical Condition Inspection (ACI) Work Specifications. 6
8. Analytical Condition Inspection (ACI) Reporting. 6
9. Interface With Other Programs. 7

Attachment 1—GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION 8

Attachment 2—SAMPLE SIZE CALCULATION 10

1. Overview. ACI programs reveal defects that may not otherwise be detected through normal technical order and programmed depot maintenance (PDM) inspections. They provide data for engineering and technical evaluations of the relative condition of the total mission, design and series (MDS) aircraft force. Data generated through ACIs is used to refine mandatory aircraft inspection programs (field and depot). Terms and abbreviations used in this instruction are listed in **Attachment 1**.

2. Responsibilities. This instruction assigns the following responsibilities:

2.1. MAJCOM:

2.1.1. Exercises surveillance over ACI programs.

2.1.2. Reviews annually (as part of the PDM program review, if one exists) proposed ACI programs and the results of existing ACI programs.

2.1.3. Approves extensions to existing ACI programs.

2.2. Air Logistics Complex (ALC). Reviews annually (as part of the PDM program review, if one exists) proposed ACI programs and the results of existing ACI programs.

2.3. Program Office (PO):

2.3.1. Sets up and monitors required ACI programs for assigned aircraft.

2.3.2. Determines ACI program requirements and develops the ACI sampling plans for assigned aerospace systems.

2.3.3. Encourages the using commands to participate in the ACI program by requesting their recommendations on ACI program requirements.

2.3.4. Establishes ACI program requirements as early as possible in each fiscal year planning cycle to facilitate program approval and implementation.

2.3.5. Prepares work specifications for the accomplishment of ACI requirements.

2.3.6. Formulates the proposed ACI program and submits for ALC and MAJCOM review (as an integral part of the PDM program package, if it exists).

2.3.7. Summarizes results of ACI programs and submits for ALC and MAJCOM review (as an integral part of the PDM package if one exists).

2.3.8. Analyzes ACI data and correlates with data from PDM, field reports, material deficiencies, accident reports and incident reports to resolve the findings.

2.3.9. Makes adjustments to technical order inspection requirements as required by analysis of ACI data.

2.3.10. Participates to the extent feasible in inspection evaluation when deficiencies are found that are considered critical.

2.3.11. Specifies the format in which ACI results must be documented and establishes procedures to ensure the priority reporting of those defects identified as major or critical.

2.3.12. Solicits inputs from the aircraft corrosion and Aircraft Structural Integrity Program (ASIP) managers regarding ACI requirements.

2.3.13. Coordinates inspection techniques for ASIP critical points with the ASIP Manager and Non-Destructive Inspection (NDI) Manager.

2.3.14. Coordinates corrosion detection, correction and prevention techniques with the corrosion manager.

2.3.15. Provides corrosion and ASIP Managers with ACI program results.

2.4. Source of Repair (Organic or Contract):

2.4.1. Accomplishes ACIs on aircraft as programmed by the PO.

2.4.2. Reports ACI results according to the format and process set up by the PO.

3. Procedures. Any item or area deemed applicable by the Program Office (PO) may be included in an ACI program. ACI programs do not cover all areas of aircraft and systems every year; rather, the inspection and evaluation process is continuous to ensure airworthiness.

3.1. Results from ASIP, failure data, and material deficiency and accident or incident reports must be the primary sources for determining which items or areas require inspection and evaluation on an annual basis. Also, highly stressed areas, ASIP identified critical points, inaccessible areas, and areas subject to corrosion, such as wheel wells, surface controls, bomb bays, and lower wing surfaces, must be carefully screened for inclusion in an annual ACI program.

3.2. A sampling of aircraft that are leading the fleet in hours, severity, or environment severities of exposure should be evaluated to identify and validate operational performance.

Proactive inspection results will provide for both timely solution development, implementation in ongoing production, economical management, and risk mitigation.

3.3. Areas such as subsystems or secondary structures are scheduled for inspection and evaluation using a phased ACI approach. For example, fuel systems and life support systems may be inspected and evaluated in one year, flight controls and hydraulic systems the next year, and avionics and secondary structures the third year.

3.4. ACI data must be used to determine the condition of the remaining MDS force. When necessary, additional data may be gathered by:

3.4.1. Sampling additional aircraft in for PDM.

3.4.2. Initiating field level inspections.

3.4.3. Evaluating existing assets in supply.

3.4.4. Accomplishing further engineering analysis.

3.5. The ACI program must be accomplished concurrently with modification and PDM program requirements.

4. Initiating Analytical Condition Inspection (ACI) Programs. The PO must determine when an ACI program should be initiated and must maintain the technical data used to support the decision.

4.1. General guidance on ACI program initiation by aircraft type. Deviations to this general guidance must be technically justified by the PO.

4.1.1. Trainer or fighter type aerospace systems: when four or more aircraft have been in the operational inventory for 3 years or have attained 1,000 flight hours of service.

4.1.2. Bomber, cargo or tanker type aerospace systems: when four or more aircraft have been in the operational inventory for 5 years or have attained 6,000 flight hours of service.

4.1.3. For other types of systems such as non-expendable Unmanned Aerial Vehicles (UAV) or long-term test assets, an appropriately tailored ACI program should be considered.

4.1.4. ACI programs are generally not required for commercial aircraft maintained under Federal Aviation Administration (FAA) requirements.

4.2. The PO must prepare a proposed ACI program for review by the air logistics complex (ALC) and major command (MAJCOM). The program must be prepared for each MDS of aircraft. Where there are significant differences in design or use, each series of aircraft must be treated as a separate grouping for gathering condition data. Of the various possible strata of an MDS (aircraft configuration, calendar age, mission, operational environment, flying hours, etc.), normally the most severe stratum is selected for the collection of basic ACI data, with discrete tasks for suspect or critical areas added from other strata to the basic data package. ACI programs should include aircraft that are being used in Controlled Interval Extension (CIE) programs. ACI groups may vary by aerospace system based on peculiarities within the particular series, historical data, and differences in PDM cycles.

4.3. The PO determines ACI groupings and quantities for review and approval by the ALC and MAJCOM. ACI data is accrued within the approved fiscal year program. The MAJCOM is the authority for extensions to the approved ACI program.

4.4. ACI programs should be considered for reduction or termination if it is determined that adequate access to areas of the aircraft is provided through planned depot level repairs or modifications. ACI programs for systems with established retirement dates should be ended as soon as practical. **(T-3)**

5. Analytical Condition Inspection (ACI) Sampling Plans. The PO must develop a sampling plan before the start of an ACI program. The plan must address the problem for which the inspections are to be conducted and must include the following:

5.1. The inspection requirements and schedule (time and place).

5.2. The sample size needed. **Note**: Scheduling or funding considerations could make it impractical to complete sampling within a single fiscal year program. In this case, the plan should explain what portion of the sample is to be completed in the current year and what portion in future years.

5.3. A definition of the anticipated defect conditions.

5.4. Any assumptions made.

5.5. A stratification of the force.

5.6. A summary of what necessary actions are projected based on the anticipated ACI findings. The sampling plan must be documented and maintained for use with later ACI analyses for the same MDS aircraft. A summary of the ACI sampling plan is included in the PDM brochure, if one exists, for review by the ALC and MAJCOM.

6. Analytical Condition Inspection (ACI) Samples. The sampling process identifies defects within a designated segment of the force, with a certain level of confidence. The sample size (the number of aircraft to be examined) varies based on the force size. A tool is provided to determine the sample size for testing from the following input values: population size (N), input defect rate (p), confidence level (CL), and margin of error (E). The tool can be accessed at this location: https://tableau.advana.data.mil/#/site/BLADE/views/ACISampleSize/Calculator

6.1. The tool calculates the sample size required for a single test and provides a Defect Rate Table to interpret the results and estimate the range of the Population Defect Rate. The last row of the table indicates the number of defects that, if found, would cause the upper bound of the range estimate to exceed the target Input Defect Rate.

6.2. To determine the smallest sample size required for a particular force size that meets the desired defect rates, confidence level, and margin of error, the tool employs a one-tailed confidence test based on the population's underlying normal distribution.

6.3. Recommendations. It is recommended to set the confidence level between 90-99%, the defect rate at 20%, and the margin of error at ±10%. While a higher confidence level enhances the precision of the sampling test, it also necessitates an increase in the margin of error for practicality. The margin of error signifies the precision of the estimate, providing a range within which the true defect rate is likely to fall with a specified level of confidence. For small fleet sizes (30 or fewer aircraft), the margin of error should be raised due to the limited sample

size that can be drawn from a small population, resulting in a larger sampling error and higher margin of error; the confidence level may also be lowered. **Note**: Statistical sampling methods may not be practical for very small fleets (8 or fewer aircraft); in such cases, the PO may devise an alternate sampling strategy – or inspect the entire fleet – and must technically justify its decision.

6.3.1. Possible confidence level (*CL*) values are limited to: 80%, 85%, 90%, 95%, 97.5%, 99% (default 90%). For small fleet sizes a recommended CL is 80 or 85%.

6.3.2. Possible margin of error (*E*) values are limited to: 5%, 10%, 15% (default 10%). For small fleet sizes a recommended margin of error is ±15%.

6.3.3. Possible input defect rate (*p*) values are limited to 1% increments in the range 5-50% (default 20%). It is not recommended to change this defect rate unless there is a specific reason to target a different threshold. Do not set *p* lower than *E*.

6.4. The sample defect rate estimates the true population defect rate but varies depending which aircraft are selected. The sample defect rate is the observed number of defects detected in the sample divided by the adjusted sample size. The estimated population defect rate ranges in the tool's Defect Rate Table are calculated in proportion to the sample defect rate using the margin of error. The confidence level represents the likelihood that the true population defect rate falls within the given range.

6.5. The sample size and selection criteria adhere to statistical probability theory, ensuring a statistically valid sample of the fleet population. The selection of ACI sample aircraft must be made through coordination with the using command or commands.

6.6. When multiple areas are being inspected as part of an ACI program, the number of defects found in each area is interpreted separately to estimate the population defect rate range of each type of defect.

7. Analytical Condition Inspection (ACI) Work Specifications. ACI work specifications are designed to identify hidden defects, deteriorating conditions, corrosion, fatigue, overstress and other deficiencies in the aircraft structure or systems. Work specifications are divided into annual and phase tasks.

7.1. Annual tasks in the work specification and task brochure must commence with two-digit task identifier composed of section and type (e.g., 2=ACI, A=Annual)

7.2. Phased tasks in the work specification and task brochure must commence with two-digit task identifier composed of section and type (e.g., 2=ACI, B=Phased)

7.3. For ACI tasks no longer required, the identification number and title must be retained on the work specification master list. The last year of accomplishment must be indicated as well as its disposition, e.g., moved to PDM, deleted, added to -6 technical order, etc.

7.4. On the work specification master list, the page on which the task is illustrated must be listed.

8. Analytical Condition Inspection (ACI) Reporting. The PO must specify the format in which ACI results are to be documented to ensure the engineering activity is provided a basis for review and evaluation.

8.1. When a defect is found during an ACI, the performing activity must initially classify the defect using one of the three following categories:

8.1.1. Minor defect. A defect that is not likely to reduce materially the usability of the aircraft for its intended purpose.

8.1.2. Major defect. A defect other than critical that is likely to result in failure or to reduce materially the usability of the aircraft for its intended purpose.

8.1.3. Critical defect. A defect that is likely to result in unsafe conditions for persons using, maintaining or depending on the aircraft, or to prevent performance of the tactical function of the aircraft.

8.2. The PO must establish procedures to ensure the priority reporting of those defects identified as major or critical.

9. Interface With Other Programs. ACI type reports of PDM inspections conducted on CIE and baseline PDM aircraft should provide data for analysis under the CIE program. Consideration should be given to combining ACI, PDM and CIE programs on the same aircraft.

BRIAN R. MOORE, Brigadier General, USAF
Director of Logistics, Civil Engineering, Force Protection, and Nuclear Integration

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

DAFPD 21-1, *Maintenance of Military Materiel,* 21 February 2024

DAFMAN 90-161, *Publishing Processes and Procedures,* 18 October 2023

AFI 33-322, *Records Management and Information Governance Program,* 23 March 2020

Prescribed Forms

None

Adopted Forms

None

Abbreviations and Acronyms

ACI—Analytical Condition Inspection

AFRIMS—Air Force Records Information Management System

ALC—Air Logistics Complex

ASIP—Aircraft Structural Integrity Program

CIE—Controlled Interval Extension

DAF—Department of the Air Force

DAFMAN—Department of the Air Force Manual

DAFPD—Department of the Air Force Policy Directive

FAA—Federal Aviation Administration

FPC—Finite Population Correction

MAJCOM—Major Command

MDS—Mission, Design, and Series

NDI—Non-Destructive Inspection

PDM—Programmed Depot Maintenance

PO—Program Office

RDS—Records Disposition Schedule

TAI—Total Active Inventory

UAV—Unmanned Aerial Vehicle

Terms

Aircraft Structural Integrity Program (ASIP). —A time-phased set of required actions performed at the optimum time during the lifecycle (design through phase-out) of an aircraft system to ensure the structural integrity (strength, rigidity, damage tolerance, durability and service life capability) of the aircraft.

Analytical Condition Inspection (ACI). —The systematic disassembly and inspection of a representative sample of aircraft to find hidden defects, deteriorating conditions, corrosion, fatigue, overstress and other deficiencies in the aircraft structure or systems. ACIs are normally over and above those inspections specified in the technical order or PDM work specifications.

Controlled Interval Extension (CIE). —The controlled extension of a PDM interval based on condition analyses of a representative sample of aircraft.

Programmed Depot Maintenance (PDM). —Depot level inspections and maintenance scheduled on a cyclic basis.

Attachment 2

SAMPLE SIZE CALCULATION

A2.1. Overview. This attachment outlines the statistical principles and theory used in the Sample Size Calculator tool for calculating sample size required when testing for defect rates in a fleet population, considering the user's desired confidence level and margin of error. This tool is based upon well-established methods for estimating sample sizes for proportions, with an adjustment for finite population sizes using the Finite Population Correction (FPC).

A2.2. Sample Size Formula and Definitions. The tool calculates sample size using the following formula:

A2.2.1. Adjusted sample size, , reflects the sample size calculated and output by the tool with the Finite Population Correction (FPC) applied. The output of the tool ensures that the sample is large enough to provide a statistically valid estimate of population defect rate.

A2.2.2. Sample size, ***n***, represents the number of aircraft needed to be tested to estimate the defect rate of the population within a specified margin of error and confidence level. This is the unadjusted sample size, which would be valid for extremely large populations

A2.2.3. The z-score, ***Z***, corresponds to the critical value based on the desired Confidence Level (CL) and indicates how many standard deviations away from the mean the sample estimate should be.

A2.2.3.1. The confidence level chosen by the user determines how much of the distribution is covered by the interval and influences the z-score used in the calculation (e.g., A confidence level of 90% means $\boldsymbol{Z} = 1.28$. If we increase the confidence level to 95%, then $\boldsymbol{Z} = 1.64$. An increase in z-score causes an increase in sample size since the higher the confidence, the larger the z-score, and the higher the sample size to be more certain that the estimate falls within the specified margin of error.)

A2.2.4. The input defect rate, ***p***, is the estimated proportion of the population that has the defect and is used to calculate the expected variability in the population as it is used to calculate sample size.

A2.2.5. The margin of error, ***E***, represents the maximum acceptable difference between the sample estimate and the true population parameter (defect rate).

A2.2.5.1. Defines the precision of the estimate; therefore, a lower margin of error would require larger sample sizes for the same level of confidence.

A2.2.6. The population size, ***N***, is the total number of aircraft in the fleet population.

A2.2.6.1. When the population is small, the tool adjusts the sample size through a finite population correction, reducing the sample size needed as a larger proportion of the population is sampled, for practicality.

A2.3. Numerator:

A2.3.1. This part of the formula calculates the unadjusted sample size, ***n***, as though the population were infinite or very large.

A2.3.2. All four of the variables are based on user-input parameters in the tool: the z-score squared, Z^2, reflects the desired confidence level, p is the defect rate (the population proportion), $(1-p)$ is the complement of the defect rate (i.e., the proportion of non-defective items) and E is the margin of error.

A2.4. Denominator:

A2.4.1. The denominator introduces the Finite Population Correction (FPC) factor, which adjusts the sample size when working with a finite population, such as a fleet of aircraft.

A2.4.2. This correction accounts for the fact that as the sample size approaches the total population size, the variability of the sample decreases because the sample becomes more representative of the entire population.

A2.4.3. N is the total population size (in this case, the total number of aircraft in the fleet).

A2.5. Interpretation of the Method.

A2.5.1. Sample Size for Infinite Population: The first part of the formula (the numerator) calculates the sample size assuming that the population is large enough that finite population effects are negligible. This part is derived from the standard formula for calculating sample size needed to estimate a population proportion with a specified confidence level and margin of error:

A2.5.2. Finite Population Correction (FPC): The second part of the formula (the denominator) introduces the finite population correction (FPC). The FPC is used when sampling from a finite population. The FPC reduces the required sample size when the population is small, because as a larger proportion of the population is sampled, the estimates become more accurate, and fewer samples are needed to achieve the same level of precision. However, when the population size is large, the FPC reduction is minimal.

A2.5.2.1. The FPC is typically applied like this:

A2.5.2.2. The tool incorporates this adjustment to the sample size, reducing it when N is small, but with a diminishing effect as the population size grows.

A2.5.3. Estimated Defect Rate Range Table. The tool also produces a table showing the estimated range of the Population Defect Rate for the number of defects detected in the sample. This range of possible defect rates for the population is generated by applying the margin of error to account for sampling variability. The estimated population defect rate is the proportion of defective aircraft in the fleet population, based on the sample tested.

A2.5.3.1. The estimated population defect rate range considers the sample size, the (hypothetical) number of defects found in the sample and the margin of error that defines the range within which the true defect rate is likely to fall. This estimate produces a range for the defect rate (e.g., “0.0% to 10.0%”), providing both the lower and upper bounds of the defect rate in the population, based on the variability inherent in sampling. Instead of relying on a single, potentially misleading figure (like an exact defect rate), the tool gives a range of possible defect rates. This range reflects the margin of error, providing a more accurate view of the population’s defect rate.

A2.5.3.2. The tool automatically adjusts the number of rows shown in this table up to the least number of defects that would cause the upper bound of the estimated range to go

above the user-selected value of ***p***. **Note**: The table does not go above 20 defects. It is possible to select parameters (***N, CL, E, p)*** that may generate impractically large Sample Size recommendations.

A2.5.3.3. The lower and upper bounds are constrained to the actual minimum and maximum possible population defect rate given the number of defects found and the number of aircraft not sampled. (That is, the width of the range estimate may be less than $2 \cdot \boldsymbol{E}$ in some cases.)

A2.5.4. The tool also provides a concluding statement for the maximum number of defects that can be found in the sample *without* causing the upper bound of the estimated range to go above the input defect rate, ***p***.

A2.6. Assumptions and Limitations. It is important to note and understand the boundaries of the tool's applicability and the conditions under which the results are valid. The purpose of this section is to clarify the key conditions under which the tool produces valid results, while also setting expectations for potential deviations based on factors like accuracy of input data, sampling methods, and distribution of defects in the population. Understanding these assumptions and limitations ensures that users can apply the tool correctly and interpret its outputs appropriately, while also recognizing scenarios where adjustments or further analysis might be needed.

A2.6.1. The tool assumes that the sample of aircraft selected for inspection is randomly chosen and representative of the entire fleet. If the sample is biased, the results may not accurately reflect the overall defect rate in the fleet population.

A2.6.2. The tool assumes that each observation in the sample is independent of the others. This may not be true if there are correlations between observations (e.g., one aircraft's defect is related to another's, due to shared parts or maintenance history).

A2.6.3. The calculation assumes that defects follow a binomial distribution (i.e., each aircraft is either defective or not). This is a typical assumption in proportion-based sample size calculations, which may not hold in cases of highly skewed or complex defect distributions.

A2.6.4. The tool calculates sample sizes and estimates based on the assumption that each aircraft is either defective or not. It does not account for cases where multiple defects could be present on the same aircraft, which might require a more detailed analysis.

A2.6.5. The formula includes the Finite Population Correction (FPC), which assumes that the population is finite and that the sample size should be adjusted as the sample size approaches the total population. The correction is most useful for smaller fleet populations but may have a negligible effect for large populations.

A2.6.6. Errors in data entry, calculation, or interpretation within the tool can affect the accuracy of the results. Users should interpret the results with caution and consider the possible limitations and sources of error.

A2.7. Interpretation of Results. The purpose of this section is to provide guidance on how to interpret the output of the tool and translate the statistical findings into actionable insights.

A2.7.1. The tool provides two main outputs: the adjusted sample size, , and the estimated population defect rate range table.

A2.7.2. The tool calculates the number of aircraft that need to be inspected as a sample to estimate the defect rate for the entire fleet. The sample size represents the minimum number of aircraft that need to be inspected for defects. The tool uses the values input by the user for the calculation: fleet population size, confidence level, margin of error, and input defect rate.

A2.7.3. The tool also provides an estimated defect rate range table that reflects the possible values of the true defect rate for the entire fleet based on the results of the sample. This range accounts for the inherent variability in sampling and the margin of error. The estimated population defect rate range is the range within which the true defect rate of the entire fleet is likely to fall, given the sample that was inspected. The range is calculated by applying the margin of error to the sample defect rate.

A2.7.3.1. For example, if a sample of 21 aircraft shows 3 defective aircraft (14.3% defective in the sample) and the margin of error is ± 10%, then the estimated defect rate range for the fleet would fall between 4.3% and 24.3%.

A2.8. Why Use This Method?

A2.8.1. This tool adheres to well-established statistical principles, ensuring that the sample size and defect rate estimates are statistically valid. The underlying formulas are derived from the binomial distribution for proportions and adjusted using the normal approximation, making the results robust and reliable. The formula used for sample size estimation is widely accepted in statistics for situations involving finite populations, where the correction ensures no oversampling when the population size is relatively small.

A2.8.2. The tool can be easily adapted for different fleet sizes and desired confidence levels. Whether the fleet population is large or small, the tool accounts for the fleet size and adjusts the required sample size accordingly for practicality. Also, the level of precision can be adjusted with the margin of error, depending on the criticality of the testing at hand.

A2.8.3. The automated calculations and clear, actionable outputs make it easy for users without a statistical background to apply the tool with complex statistical concepts correctly, while understanding the outputs of the tool for sample size and estimated population defect rate range. The tool provides a range of possible population defect rates, which is a more accurate way to represent the uncertainty that occurs with sampling. This tool gives a better understanding of the potential outcomes and provides a stronger foundation for informed decision-making in managing fleet quality and maintenance.